

Week 1 HTML

COSC3060 - Web Programming Studio

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Recall: programming

What is programming?

- To write computer instructions to make computer programs

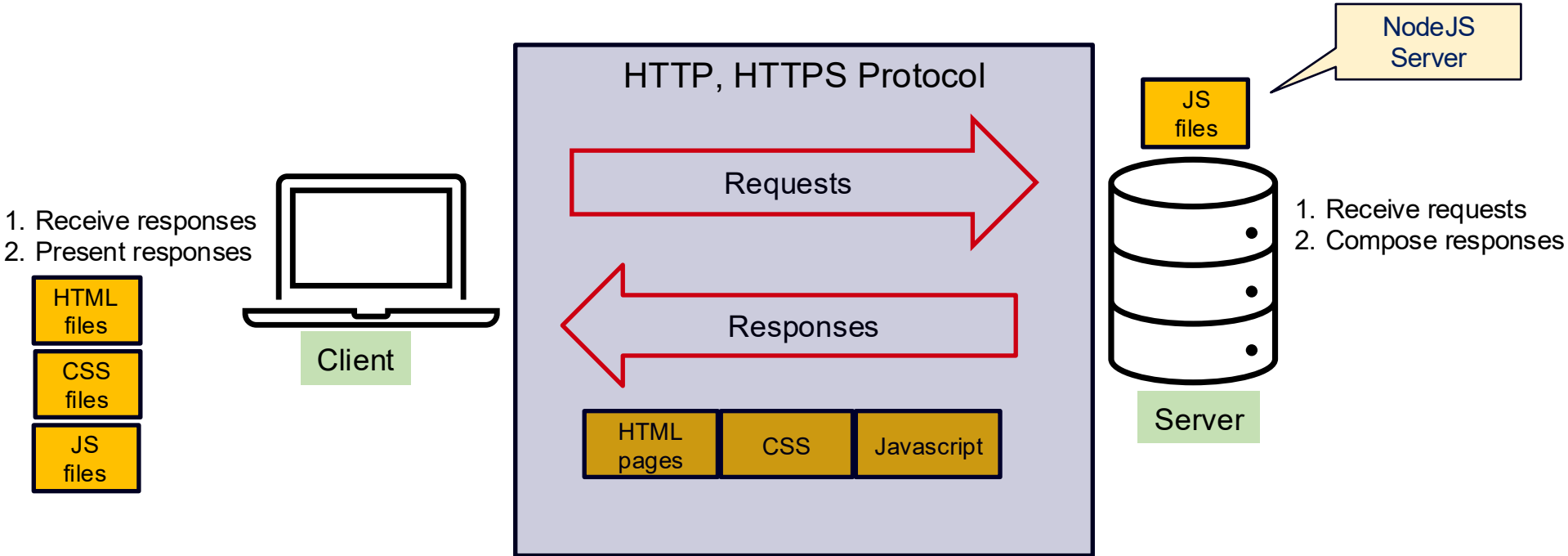
What is web programming?

- To write computer instructions to make web applications.
- Web applications follow client-server architecture, we learn about client-server architecture in order to program web applications.

Fundamentals about web programs

- How the web system operates
 - web browser
 - web server
 - DNS
 - Popular internet protocols
- How web standards model
- How browsers load websites

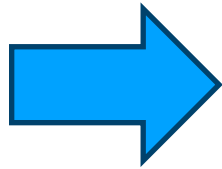
Client-server architecture



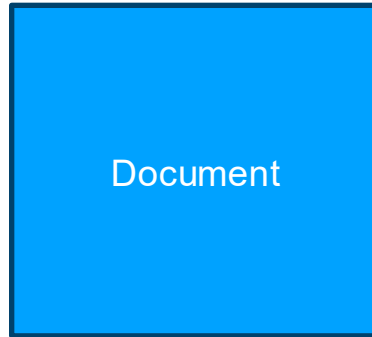
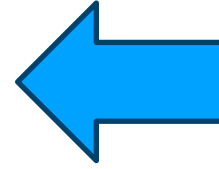
HTML, CSS, JAVASCRIPT

Basis languages to make webpages are HTML, CSS, JAVASCRIPT

HTML structures



CSS decorates



Javascript animates



Lesson outline

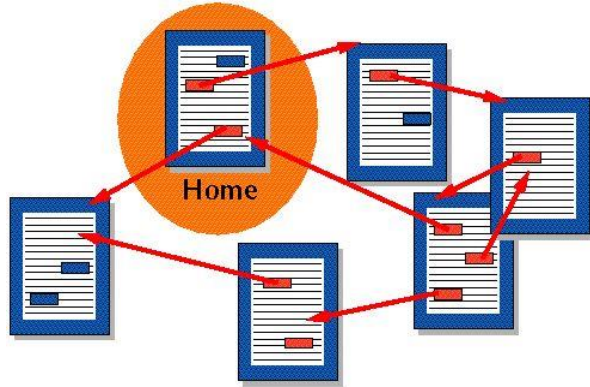
1. HyperText Markup Language (HTML)

- Anatomy of an HTML element
- HTML Tag Attributes
- Basic structure of HTML document
- Some basic HTML Tags

2. Hosting with Github Pages

HTML = Hyper Text Markup Language

- **Hyper Text** = Text with connection from/to other text
- **Markup language** : communicate by bringing structure and semantic to text
- **HTML** describes structure of a hypertext and its connection to others
- **Hypertext** : a line of text, a document, a webpage, a website



HTML

- a markup language to define content and structure of webpages
- makes digital content to be machine-readable, searchable, interconnected
- makes webpages by using HTML elements
- Syntax:

```
<!DOCTYPE html>
<html>
  <head>
    ... HTML elements
  </head>
  <body>
    ... HTML elements
  </body>
</html>
```



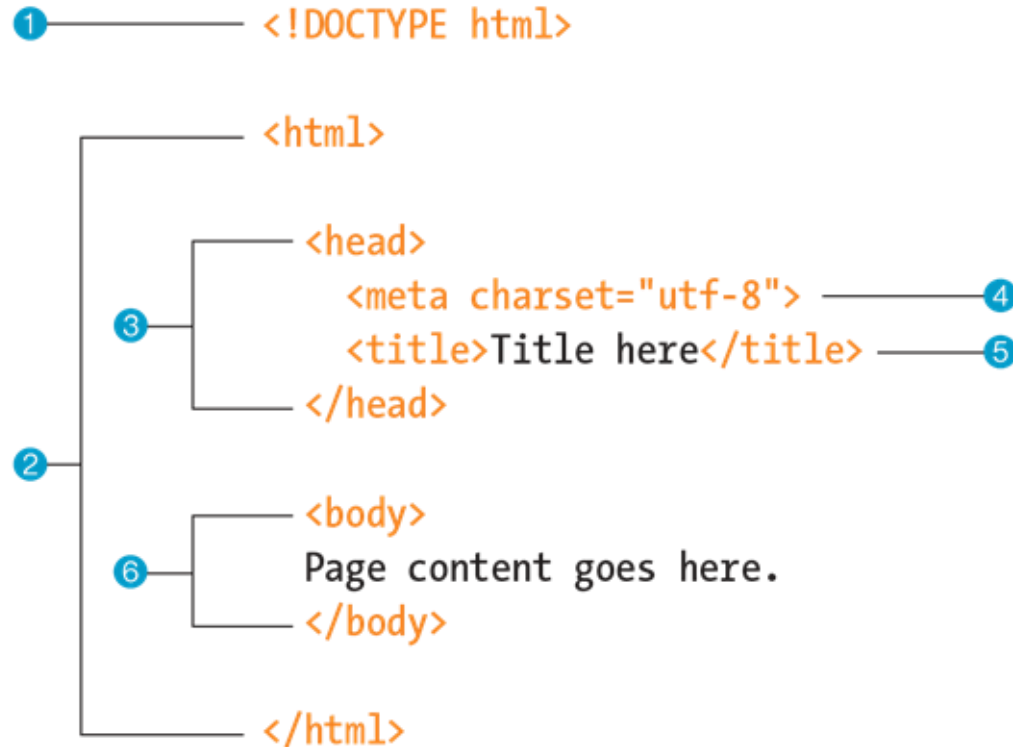
Unit of an HTML
webpage
HTML Element

HTML = “Vocabulary” + “Grammar”

- HTML Elements
- Structure of an HTML document
- HTML tag attributes

Basic structure of HTML document

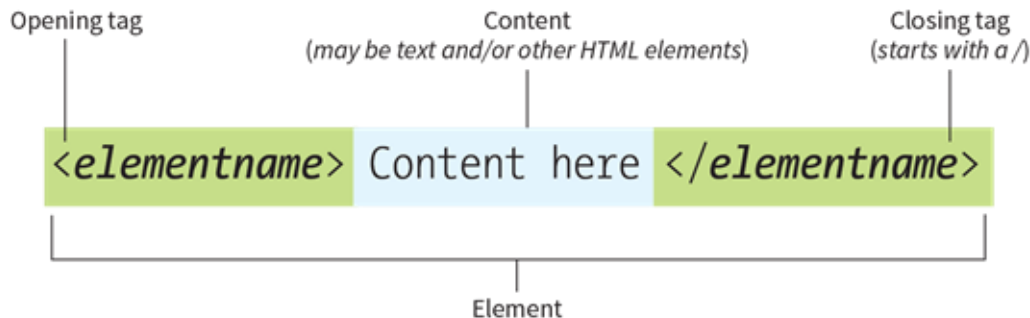
1. **DOCTYPE** tells browsers which HTML version to use.
2. **HTML** element is the root and contains all other elements.
3. **Head** and **body** elements divide the document.
4. **Meta** elements provide document metadata, including character encoding.
5. **Title** element is required and provides a descriptive title.
6. **Body** element contains content to display in the browser.



Anatomy of an HTML element <> </>

An HTML element consists of several parts, including:

- **Opening tag:** starts an HTML tag, consists of the element name surrounded by < >
- **Content:** the information that appears between the opening and closing tags.
- **Closing tag:** ends an HTML tag and has identical element name to the opening tag, includes a forward slash before the element name.



Example:

`<h1>`Black Goose Bistro`</h1>`

HTML Tag Attributes

additional properties added to an HTML tag to provide extra information about the tag's content or behavior.





Let's dive deeper with some basic HTML elements



- Headings
- Paragraph
- List
- Anchor
- Multimedia
- Table

<h1> to <h6> tags (Heading elements)

The **<h1>** to **<h6>** HTML elements represent six levels of section headings.

<h1> is the highest section level and **<h6>** is the lowest.

```
<h1>Heading level 1</h1>
```

```
<h2>Heading level 2</h2>
```

```
<h3>Heading level 3</h3>
```

```
<h4>Heading level 4</h4>
```

```
<h5>Heading level 5</h5>
```

```
<h6>Heading level 6</h6>
```

```
<h2>This is a header created using the h2 tag</h2>
```

Heading level 1

Heading level 2

Heading level 3

Heading level 4

Heading level 5

Heading level 6

This is a header created using the h2 tag

<p> tag (Paragraph element)

- The `<p>` HTML element represents a paragraph.

```
<p>  
  This is the first paragraph of text. This is the first paragraph of text. This  
  is the first paragraph of text. This is the first paragraph of text.  
</p>  
<p>  
  This is the second paragraph. This is the second paragraph. This is the second  
  paragraph. This is the second paragraph.  
</p>
```



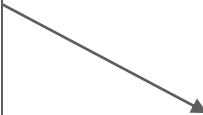
This is the first paragraph of text. This is the first paragraph
of text. This is the first paragraph of text. This is the first
paragraph of text.

This is the second paragraph. This is the second paragraph.
This is the second paragraph. This is the second paragraph.

 tag (Line Break element)

- The **
** HTML element produces a line break in text (add a new line).

```
<p>Here is my address:</p>
<p>
  123 Nguyen Van Linh street<br>
  Tan Hung ward<br>
  District 7<br>
  Saigon
</p>
```



Here is my address:
123 Nguyen Van Linh street
Tan Hung ward
District 7
Saigon

Question. Can you notice of the break space difference between two **<p>** elements and **
** tags?

<hr> tag (Horizontal Rule)

The `<hr>` HTML element represents a thematic break between paragraph-level elements

```
<p>First rule: The first rule of Fight Club is: You do  
not talk about Fight Club.</p>  
<hr>  
<p>Second rule: The second rule of Fight Club is: Always  
bring cupcakes.</p>
```

First rule: The first rule of Fight Club is: You do not talk about Fight Club.

Second rule: The second rule of Fight Club is: Always bring cupcakes.

[Question] What is the special attribute of `<hr>` tag to adjust the thickness of the line? (hint: check MDN)

Example with some HTML tags

- With a few basic HTML tags, you can start to organize content in some logical and neat ways and sections.

```
<h1>Heading elements</h1>
<h2>Summary</h2>
<p>Some text here...</p>
<hr>
<h2>Examples</h2>
<h3>Example 1</h3>
<p>Some text here...</p>
<hr>
<h3>Example 2</h3>
<p>Some text here...</p>
<hr>
<h2>See also</h2>
<p>Some text here...</p>
```

Heading elements

Summary

Some text here...

Examples

Example 1

Some text here...

Example 2

Some text here...

See also

Some text here...

Comments in HTML

- In HTML, comments are used to add notes or explanations about a piece of code that are not meant to be displayed on the webpage.
- Comments start with `<!--` and end with `-->`.

```
<h1>Welcome to my website!</h1>
<!-- This is a comment about the main headline -->
<h2>About Me</h2>

<!-- This is a comment about the first paragraph -->
<p>I'm a web developer with 5 years of experience.</p>

<!-- This is a comment about the second paragraph -->
<p>My skills include HTML, CSS, and JavaScript.</p>

<!-- This is a comment about the third paragraph -->
<p>Feel free to contact me for any web development needs!</p>

<!-- This is a comment about the copyright information -->
<p>Copyright © 2023 My Website</p>
```

Welcome to my website!

About Me

I'm a web developer with 5 years of experience.

My skills include HTML, CSS, and JavaScript.

Feel free to contact me for any web development needs!

Copyright © 2023 My Website

 tag (Unordered List element)

The `<ul>` HTML element represents an unordered list of items, typically rendered as a bulleted list.

```
<p>My coding routine:</p>
<ul>
  <li>Wake up</li>
  <li>Coffee</li>
  <li>Code</li>
  <li>Google error messages
    <ul>
      <li>Find solutions on stack overflow</li>
      <li>Read through 200+ comments</li>
      <li>Realize none of them apply to your issue</li>
    </ul>
  </li>
  <li>Cry</li>
  <li>Repeat</li>
</ul>
```

My coding routine:

- Wake up
- Coffee
- Code
- Google error messages
 - Find solutions on stack overflow
 - Read through 200+ comments
 - Realize none of them apply to your issue
- Cry
- Repeat

[Question] Can you customize or select how the bulleted points look like instead of just boring circle? (hint: check MDN)

 tag (Ordered List element)

The **** HTML element represents an ordered list of items — typically rendered as a numbered list.

```
<h2>Blueberry Muffins</h2>
<p>These muffins are a delicious way to start your day!</p>
<h3>Ingredients:</h3>
<ol>
  <li>sugar</li>
  <li>teaspoon salt</li>
  <li>egg</li>
  <li>vegetable oil</li>
  <li>fresh blueberries</li>
</ol>
<h3>Instructions:</h3>
<ol>
  <li>Preheat oven to 400°F (200°C).</li>
  <li>In a large bowl, mix together flour, sugar, baking powder, and salt.</li>
  <li>Pour the wet ingredients into the dry ingredients and stir until just combined.</li>
  <li>Bake for 20-25 minutes, or until a toothpick inserted into the center of a muffin comes out clean.</li>
</ol>
```

Blueberry Muffins

These muffins are a delicious way to start your day!

Ingredients:

1. sugar
2. teaspoon salt
3. egg
4. vegetable oil
5. fresh blueberries

Instructions:

1. Preheat oven to 400°F (200°C).
2. In a large bowl, mix together flour, sugar, baking powder, and salt.
3. Pour the wet ingredients into the dry ingredients and stir until just combined.
4. Bake for 20-25 minutes, or until a toothpick inserted into the center of a muffin comes out clean.

[Question] What attributes of **** tag customize how the ordered list should look like instead of just numbering from 1 to n?

<a> tag (Anchor “hyperlink” element)

- The **<a>** HTML element (or anchor element), with its **href** attribute, creates a hyperlink to web pages, files, email addresses, locations in the same page, or anything else a URL can address.

<p>You can reach Tom at:</p>

SSET Website

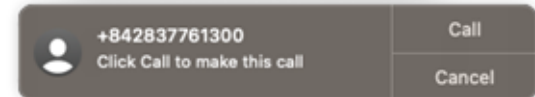
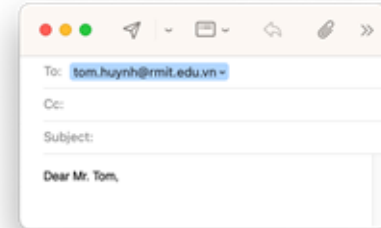
Email

Phone

[Question] The above anchor link for the website will replace your current website with the SSET website instead of opening it in a new tab, can you figure out why and how? (Hint: check MDN website)

You can reach Tom at:

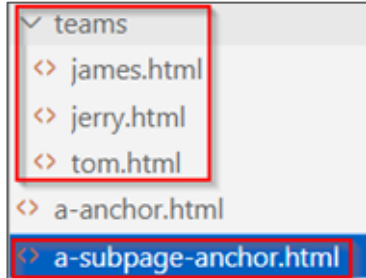
- [SSET Website](https://www.rmit.edu.vn)
- [Email](mailto:tom.huynh@rmit.edu.vn)
- [Phone](tel:+842837761300)



<a> tag (Linking to another local HTML file)

anchor tag can also link (hyperlink) to another local HTML file in a different folder like so:

Folder structure:



a-subpage-anchor.html

```
<p>Here is the list of my teammates and their profile pages:</p>
<a href="teams/tom.html">Tom</a><br>
<a href="teams/jerry.html">Jerry</a><br>
<a href="teams/james.html">James</a><br>
```



 tag (Image Embed element)

The **** HTML element embeds an image into the document.

```
<h1>My favourite cat</h1>  

```



↓ ✗

My favourite cat

 a famous Vietnamese cat image on Tiktok

[Explanation] When tag cannot load the image then the alt text will be shown instead.

[Question] Which img attributes resize the image? (Hint: check MDN website)

<table> tag (Table element)

- <table>
 - <caption> Content </caption>
 - <thead>
 - <tr>
 - <td>Column</td>
 - <td>Column</td>
 - </tr>
 - </thead>
 - <tbody>
 - <tr>
 - <td>Column content</td>
 - <td>Column content</td>
 - </tr>
 - </tbody>
- </table>

The <video> element with attributes

```
<video controls
  width="400" height="400"
  autoplay
  loop
  muted
  preload="auto"
  poster="video-cover.png" >
  <source src="video-file.mp4" type="video/mp4" />
  <source src="video-file.webm" type="video/webm" />
  <p> Your browser doesn't support this video.
    Here is a <a href="video-file.mp4">link to the video</a> instead.
  </p>
</video>
```

The <audio> element

<audio controls>

<source src="viper.mp3" type="audio/mp3" />

<source src="viper.ogg" type="audio/ogg" />

<p> Your browser doesn't support this audio file.

Here is a link to the audio instead.

</p>

</audio>

HTML Semantic elements

HTML elements should convey not only **presentation** of a text but also **meaning** of it.

Benefits from semantic HTML:

- Web pages become effectively searchable
- More readable and helpful for visually impaired users
- Code more understandable for developers

Almost all HTML elements are semantics, except `<div>`, `<span>`, `<section>` being generic

HTML style attribute

Special attribute to style or decorate HTML elements

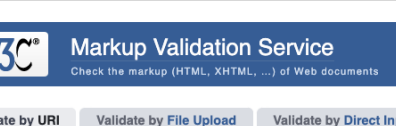
Syntax

```
<elementname style="attr1: val; attr2: val; ... ">  
</elementname>
```

Debugging HTML

Check an HTML page for markup errors

- Your understanding about HTML
- Tool
 - HTML validator <https://validator.w3.org/>
 - DOM Inspector in the browser



https://validator.w3.org

W3C[®] Markup Validation Service

Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI **Validate by File Upload** **Validate by Direct Input**

Validate by URI

Validate a document online:

Address:

[More Options](#)

Check



```
<!DOCTYPE html>
<html lang="en-US" prefix="og: https://ogp.me/ns#" class="os-default">
  <head>
  </head>
  <body>
    <script>
    </script>
    <div id="root">
      <ul id="nav-access" class="ally-nav">
      </ul>
      <div class="page-wrapper category-learn document-page">
        <div class="top-banner visible" style="--place-top-background-light: #FFFFFF;
        --place-top-cta-background-light: #016680; --place-top-cta-color-light: #1
        dk: #FFFFFF; --place-top-color-dark: #000000; --place-top-cta-background-dark
        -dark: #FFFFFF;">
        </div>
        <div class="sticky-header-container">
        </div>
        <div class="main-wrapper">
          <div class="sidebar-container">
            <aside id="sidebar-quicklinks" class="sidebar">
            </aside>
            <div class="toc-container">
            </div>
            <div class="toc">
            </div>
          <div class="place side">
          </div>
        </div>
      </div>
      <div id="content" class="main-content">
      </div>
      <div class="bottom-banner-container" style="--place-banner-width: 728px; --p
```

